

AWS Service Dictionary for AIF-C01

Complete Service Reference Guide

AWS AI Cloud Practitioner Certification Prep Program - Task 7 of 15 Based on AIF-C01 Exam Guide Version 1.1 (April 30, 2026) and the Master Project Blueprint

How to Use This Dictionary

This dictionary is organized by AWS service category, and alphabetically within each category. Every entry uses the same seven-field format so you can scan quickly: what the service is, an everyday analogy, its primary use case, why it matters for the exam, a short business scenario, the phrases that should make you think of this service on exam day, and the service it gets confused with most often.

Two important labels appear throughout this guide:

- **Officially In-Scope** - This service appears on the AWS-published AIF-C01 Exam Guide’s in-scope services list (directly, or as a documented sub-feature of an in-scope service). Treat these as fair game for the exam.
- **Supplemental (Not Officially In-Scope)** - This service does **not** appear on the AWS-published in-scope services list. It is included here only because it is commonly mentioned alongside AI/ML topics or appears in related AWS training material, and understanding it may build helpful context. **Do not expect this service to be directly tested.** Treat any “priority” or “exam clue” language in these entries as general AWS literacy, not AIF-C01 exam guidance.

Use the two summary tables at the end of this document for rapid-fire review the night before your exam.

SECTION 1: MACHINE LEARNING AND AI SERVICES

Amazon SageMaker AI

Category: Machine Learning **Exam Priority:** High **Domain(s):** Domain 1, Domain 3, Domain 4 **What It Is (Plain English):** Amazon SageMaker AI is AWS’s full-service platform for building, training, and deploying your own custom machine learning models, covering everything from raw data to a live model running in production. **Everyday Analogy:** SageMaker AI is like a fully equipped commercial kitchen: it gives you every appliance and tool you need to create a dish from scratch, rather than handing you a finished meal. **Primary Use Case:** Organizations that need a custom-trained model (not just an existing foundation model) use SageMaker AI to manage the entire model-building lifecycle in one place. **Why the AI Practitioner Needs to Know It:** The exam repeatedly contrasts SageMaker AI (“build your own model”) against Amazon Bedrock (“use an existing foundation model”). You need to recognize which one fits a given scenario and know SageMaker AI’s role across the AI/ML pipeline. **Business Scenario:** A logistics company wanted to predict delivery delays using its own historical shipment data. They used Amazon SageMaker AI to clean the data, train a custom regression model, and deploy it as a real-time endpoint. The

result was delay predictions tailored precisely to their unique shipping network, something no off-the-shelf foundation model could replicate. **Exam Clue Phrases:** - “Build a custom model from scratch” - “Train a model on our own proprietary data” - “Full control over the ML pipeline” - “Data scientists need an end-to-end ML environment” - “Hyperparameter tuning” or “feature engineering” **Commonly Confused With:** Amazon Bedrock - SageMaker AI is for building and training your own models; Bedrock is for accessing and customizing pre-built foundation models via API without managing infrastructure.

Key Amazon SageMaker AI Sub-Features

Sub-Feature	What It Does	Exam Relevance
SageMaker Studio	A unified, web-based IDE for the entire ML workflow (notebooks, training, deployment)	Know it’s the central workspace; not usually tested on UI details
SageMaker Canvas	A no-code, visual interface that lets business users build ML models without writing code	Know it as the “no-code” option for non-technical users
SageMaker Autopilot	Automated machine learning (AutoML) that automatically builds, trains, and tunes models	Know it as SageMaker’s AutoML feature
SageMaker JumpStart	A curated library of pre-built models and solution templates (also a standalone in-scope service - see its own entry below)	Cross-reference with the standalone JumpStart entry
SageMaker Pipelines	A workflow orchestration tool for automating and repeating ML pipeline steps	Connects to MLOps concepts in Domain 1
SageMaker Ground Truth	A data labeling service that uses human annotators (with optional ML assistance) to label training data	Connects to “labeled data” concepts in Domain 1
SageMaker Feature Store	A central repository for storing, sharing, and reusing ML features	Know it manages “features” used in training
SageMaker Model Monitor	Continuously monitors deployed models for data drift and performance degradation	Tested in both Domain 1 (MLOps) and Domain 4 (responsible AI monitoring)
SageMaker Model Registry	Catalogs and version-controls trained models for governance and deployment tracking	Connects to MLOps and governance concepts
SageMaker Clarify	Detects bias in datasets and models, and explains model predictions	Heavily tested in Domain 4 (Responsible AI)

Sub-Feature	What It Does	Exam Relevance
SageMaker Data Wrangler	A visual tool for preparing and transforming data for ML	Connects to “data preprocessing” stage of the ML pipeline
SageMaker Debugger	Identifies training problems like vanishing gradients or overfitting in near real time	Lower exam priority; know it exists for troubleshooting training

Amazon SageMaker JumpStart

Category: Machine Learning **Exam Priority:** High **Domain(s):** Domain 1, Domain 2, Domain 3 **What It Is (Plain English):** SageMaker JumpStart is a feature within SageMaker AI that gives you a curated library of pre-built ML models, including some foundation models, that can be deployed with minimal setup. **Everyday Analogy:** JumpStart is like a meal-kit service: most of the prep work is already done for you, but you are still doing some of the cooking yourself within the SageMaker kitchen. **Primary Use Case:** Teams that want a faster start than building a model from zero, but who are working within the SageMaker AI environment, use JumpStart’s templates and pre-trained models. **Why the AI Practitioner Needs to Know It:** The exam distinguishes JumpStart (FMs and templates accessed through SageMaker) from Bedrock (the dedicated foundation model platform with the broadest model selection). **Business Scenario:** A retail analytics team wanted to experiment with a foundation model without leaving their existing SageMaker workflows. They used SageMaker JumpStart to deploy a pre-trained model in a few clicks. The result was a faster proof-of-concept without having to provision and configure a model from scratch. **Exam Clue Phrases:** - “Pre-built model templates within SageMaker” - “Deploy a model with minimal setup” - “Solution templates” - “Quick-start ML solutions” **Commonly Confused With:** Amazon Bedrock - JumpStart offers select FMs through the SageMaker interface; Bedrock is the dedicated managed FM platform with a broader range of providers and models.

Amazon Bedrock

Category: Machine Learning **Exam Priority:** High **Domain(s):** Domain 2, Domain 3, Domain 4, Domain 5 **What It Is (Plain English):** Amazon Bedrock is AWS’s primary platform for accessing, customizing, and deploying foundation models from multiple AI companies through a single managed API, without managing any servers. **Everyday Analogy:** Bedrock is like a food court: you don’t cook anything yourself, you simply choose from a variety of “restaurants” (model providers) and Bedrock serves up the dish through one consistent counter (API). **Primary Use Case:** Organizations building generative AI applications, such as chatbots, summarizers, or content generators, use Bedrock to access foundation models without training anything from scratch. **Why the AI Practitioner Needs to Know It:** Bedrock is the single most heavily referenced service across Domains 2, 3, 4, and 5. Most GenAI exam scenarios funnel back to a Bedrock feature. **Business Scenario:** An insurance company wanted to launch a customer-facing claims chatbot without hiring an ML engineering team. They used Amazon Bedrock to access a foundation model, ground it in their policy documents, and apply safety controls. The result was a production-ready chatbot launched in weeks instead of months. **Exam Clue Phrases:** - “Access

foundation models from multiple providers” - “No infrastructure to manage” - “Without training a model from scratch” - “Choose between different foundation models” - “Single API for generative AI” **Commonly Confused With:** Amazon SageMaker AI - Bedrock is for using pre-built FMs through API calls; SageMaker AI is for building and training custom models from scratch.

Key Amazon Bedrock Sub-Features

Sub-Feature	What It Does	Exam Relevance
Bedrock Knowledge Bases	Connects an FM to your company’s own documents so responses are grounded in your data (RAG)	Central to Domain 3’s RAG content - high priority
Bedrock Agents	Lets a foundation model take multi-step autonomous actions and call external tools and APIs	Central to agentic AI concepts in Domains 2 and 3
Bedrock Guardrails	Sets safety limits on what a model can say or do (content filters, topic restrictions, grounding checks)	Heavily tested in Domains 4 and 5
Bedrock Model Evaluation	A managed service for assessing FM quality using automated metrics or human evaluators	Central to Domain 3’s evaluation content
Bedrock Fine-Tuning	Lets customers adapt a foundation model to a specific task using their own labeled data	Connects to Domain 3’s customization tradeoffs content

Amazon Bedrock AgentCore

Category: Machine Learning **Exam Priority:** High **Domain(s):** Domain 2, Domain 3, Domain 5 **What It Is (Plain English):** Amazon Bedrock AgentCore is a managed service for building, deploying, securing, and scaling production-grade AI agents that can take actions, call tools, and orchestrate multi-step workflows. **Everyday Analogy:** If Strands Agents is the blueprint for building a robot, AgentCore is the factory floor, security system, and maintenance crew that lets you run that robot safely at scale. **Primary Use Case:** Organizations that have built agentic AI applications use AgentCore to deploy, secure, and operate those agents reliably in a live production environment. **Why the AI Practitioner Needs to Know It:** AgentCore is the managed production counterpart to the open-source Strands Agents framework, and it introduces AgentCore Identity and AgentCore Policy as security features tested in Domain 5. **Business Scenario:** A travel booking company built an AI agent that could search flights, compare prices, and complete a booking. They used Amazon Bedrock AgentCore to deploy that agent securely with proper identity controls. The result was a production agent that could act on a customer’s behalf without exposing sensitive credentials. **Exam Clue Phrases:** - “Securely deploy AI agents in production” - “Scale agentic workflows” - “AgentCore Identity” or “AgentCore Policy” - “Manage and

orchestrate multi-step agent actions” **Commonly Confused With:** Strands Agents - AgentCore is the managed runtime for deploying and securing agents in production; Strands Agents is the open-source framework used to build those agents in the first place.

Amazon Nova

Category: Machine Learning **Exam Priority:** High **Domain(s):** Domain 2, Domain 3 **What It Is (Plain English):** Amazon Nova is Amazon’s own family of foundation models, available through Amazon Bedrock, including text, image, and multi-modal models built for high performance and cost-efficiency. **Everyday Analogy:** If Bedrock is the food court, Nova is the “house restaurant” built and run by the food court’s owner (Amazon) rather than an outside vendor. **Primary Use Case:** Organizations that want an Amazon-built, cost-efficient foundation model option (rather than a third-party model) select Nova within Bedrock. **Why the AI Practitioner Needs to Know It:** The exam may ask you to distinguish “Nova” (a specific model family) from “Bedrock” (the platform that hosts Nova and other providers’ models). **Business Scenario:** A media company wanted a cost-efficient model for summarizing large volumes of articles. They used Amazon Nova through Bedrock to handle the summarization workload. The result was lower per-token costs compared to some third-party model options, without sacrificing output quality. **Exam Clue Phrases:** - “Amazon’s own foundation models” - “Cost-efficient model family” - “Available through Bedrock” - “Multi-modal Amazon-built model” **Commonly Confused With:** Amazon Bedrock (the platform) - Nova refers to specific Amazon-built foundation models; Bedrock is the platform through which Nova and many third-party models are all accessed.

Amazon Augmented AI (Amazon A2I)

Category: Machine Learning **Exam Priority:** High **Domain(s):** Domain 4 **What It Is (Plain English):** Amazon A2I adds human review workflows into ML prediction pipelines, automatically routing a model’s output to a human expert when the model’s confidence is low. **Everyday Analogy:** A2I works like a self-checkout machine at a grocery store: most transactions go through automatically, but anything unusual gets flagged for a human attendant to verify. **Primary Use Case:** Organizations using ML for sensitive or high-stakes decisions (loan approvals, content moderation, medical document review) use A2I to add a human safety net for uncertain predictions. **Why the AI Practitioner Needs to Know It:** A2I is a core Domain 4 tool representing “human-in-the-loop” responsible AI practices, and it is a top distractor pairing with SageMaker Model Monitor. **Business Scenario:** A financial services firm used an ML model to flag potentially fraudulent transactions. They used Amazon A2I to route any low-confidence fraud predictions to a human fraud analyst before any account action was taken. The result was fewer false positives reaching customers while still catching genuine fraud quickly. **Exam Clue Phrases:** - “Low-confidence predictions need human review” - “Human-in-the-loop workflow” - “Route outputs to a human reviewer” - “Add a human checkpoint before acting on a prediction” **Commonly Confused With:** SageMaker Model Monitor - A2I routes individual low-confidence predictions to human reviewers in real time; Model Monitor automatically tracks and alerts on overall model performance degradation over time.

Amazon Comprehend

Category: Machine Learning **Exam Priority:** High **Domain(s):** Domain 1 **What It Is (Plain English):** Amazon Comprehend is a natural language processing (NLP) service that analyzes text to detect sentiment, identify entities (names, dates, places), classify topics, and extract key phrases. **Everyday Analogy:** Comprehend is like a skilled reader who can skim a stack of letters and instantly tell you the tone of each one, who is mentioned, and what they're mainly about. **Primary Use Case:** Businesses use Comprehend to automatically analyze customer reviews, support tickets, or social media posts for sentiment and key topics at scale. **Why the AI Practitioner Needs to Know It:** Comprehend is the go-to answer for "analyze the meaning of existing text" scenarios in Domain 1, and is frequently contrasted with Textract (extracting text) and Translate (converting language). **Business Scenario:** A retail chain wanted to understand how customers felt about a new product launch based on thousands of online reviews. They used Amazon Comprehend to automatically detect sentiment and extract recurring complaint themes. The result was a clear, data-backed view of customer reaction without manually reading every review. **Exam Clue Phrases:** - "Detect sentiment in text" - "Extract key phrases or entities from text" - "Analyze customer feedback at scale" - "Classify text by topic" **Commonly Confused With:** Amazon Textract - Comprehend analyzes the meaning and content of text that is already in text form; Textract extracts printed or handwritten text from images and scanned documents.

Amazon Kendra

Category: Machine Learning **Exam Priority:** Medium **Domain(s):** Domain 3 **What It Is (Plain English):** Amazon Kendra is an intelligent enterprise search service powered by ML that lets employees search across multiple data sources using natural language and get precise answers, not just a list of links. **Everyday Analogy:** Kendra is like an expert reference librarian who, instead of pointing you to a shelf of books, reads the books for you and hands you the exact answer. **Primary Use Case:** Organizations with large internal document repositories use Kendra to power an internal "ask a question, get an answer" search experience for employees. **Why the AI Practitioner Needs to Know It:** Kendra often appears as a distractor against Amazon OpenSearch Service in RAG-related questions; knowing the difference is essential for Domain 3. **Business Scenario:** A large enterprise had thousands of internal policy documents scattered across multiple systems. They used Amazon Kendra to let employees ask plain-language questions and get direct answers pulled from those documents. The result was faster access to policy information and fewer support tickets asking "where do I find...". **Exam Clue Phrases:** - "Intelligent enterprise search" - "Natural language search across documents" - "Out-of-the-box NLP search, no setup of a vector database" - "Employees need to find answers, not just documents" **Commonly Confused With:** Amazon OpenSearch Service - Kendra is a purpose-built, ready-to-use intelligent search service; OpenSearch is a flexible, general-purpose search and analytics engine that is often used to store vectors for custom RAG applications.

Amazon Lex

Category: Machine Learning **Exam Priority:** High **Domain(s):** Domain 1 **What It Is (Plain English):** Amazon Lex is a service for building conversational interfaces, chatbots and voice bots, that can understand natural language, powered by the same technology behind Amazon Alexa.

Everyday Analogy: Lex is like hiring a customer service agent who already knows how to hold a conversation; you just need to teach them your company’s specific information. **Primary Use Case:** Businesses use Lex to build custom-branded chatbots and voice assistants for customer service, ordering systems, or internal helpdesks. **Why the AI Practitioner Needs to Know It:** Lex is a Domain 1 “traditional AI service” and is frequently paired against Amazon Q (an AI assistant for internal AWS/enterprise use) as a contrast point. **Business Scenario:** A food delivery startup wanted a chatbot that could take orders through both their app and a phone line. They used Amazon Lex to build a single conversational bot that understood natural speech and text. The result was a consistent ordering experience across multiple channels without separate development efforts. **Exam Clue Phrases:** - “Build a custom chatbot or voice bot” - “Conversational interface for customers” - “Same technology as Alexa” - “Understand natural language for a custom application” **Commonly Confused With:** Amazon Polly - Lex understands and responds to conversational input (speech-to-intent); Polly converts written text into spoken audio (text-to-speech). They serve different purposes within a voice application.

Amazon Personalize

Category: Machine Learning **Exam Priority:** Medium **Domain(s):** Domain 1 **What It Is (Plain English):** Amazon Personalize uses ML to build real-time personalized recommendation systems, similar to the engines behind “customers also bought” or “recommended for you” features. **Everyday Analogy:** Personalize is like a personal shopper who has studied your past purchases and instantly knows what else you’d probably like. **Primary Use Case:** E-commerce, media, and streaming companies use Personalize to power product, content, or media recommendations tailored to each individual user. **Why the AI Practitioner Needs to Know It:** Personalize is the standard exam answer whenever a question describes a “recommendation engine” use case in Domain 1. **Business Scenario:** A streaming video service wanted to increase watch time by showing viewers content they were more likely to enjoy. They used Amazon Personalize to analyze viewing history and generate individualized recommendations. The result was longer average session times and improved customer retention. **Exam Clue Phrases:** - “Personalized product or content recommendations” - “Customers also bought” - “Recommendation engine based on user behavior” - “Real-time personalization” **Commonly Confused With:** Amazon Comprehend - Personalize generates recommendations based on user behavior data; Comprehend analyzes the content and sentiment of text, which is a different type of input entirely.

Amazon Polly

Category: Machine Learning **Exam Priority:** High **Domain(s):** Domain 1 **What It Is (Plain English):** Amazon Polly is a text-to-speech service that converts written text into natural-sounding spoken audio, supporting many languages and voice styles, including neural voices. **Everyday Analogy:** Polly is like a professional voice actor who can read any script you give them aloud, instantly and in any of dozens of voices. **Primary Use Case:** Businesses use Polly to add spoken audio to applications, such as reading articles aloud, powering IVR phone systems, or narrating e-learning content. **Why the AI Practitioner Needs to Know It:** Polly and Transcribe are a classic “opposite direction” exam pairing (text-to-speech versus speech-to-text), and this distinction is a near-guaranteed exam topic. **Business Scenario:** A news publisher wanted to

offer an audio version of every article without hiring voice talent. They used Amazon Polly to automatically convert article text into natural-sounding narration. The result was a new “listen to this article” feature delivered at a fraction of the cost of human voiceover work. **Exam Clue Phrases:** - “Convert text into spoken audio” - “Text-to-speech” - “Natural-sounding narration” - “Read content aloud automatically” **Commonly Confused With:** Amazon Transcribe - Polly converts text into speech; Transcribe converts spoken audio into text. These are opposite directions of the same general capability.

Amazon Rekognition

Category: Machine Learning **Exam Priority:** High **Domain(s):** Domain 1 **What It Is (Plain English):** Amazon Rekognition is a computer vision service that analyzes images and video to detect and label objects, faces, text, scenes, activities, and inappropriate content. **Everyday Analogy:** Rekognition is like a security guard watching a video feed who can instantly tell you everything visible in the frame: who is there, what objects are present, and whether anything looks unsafe. **Primary Use Case:** Organizations use Rekognition for use cases like content moderation, identity verification, and searchable media libraries based on what appears in images or video. **Why the AI Practitioner Needs to Know It:** Rekognition is the standard “computer vision” answer in Domain 1 and is frequently contrasted against Textract, which extracts text rather than analyzing visual scene content. **Business Scenario:** A social media platform needed to automatically flag inappropriate images before they were posted publicly. They used Amazon Rekognition to scan uploaded images and detect explicit or violent content in real time. The result was faster content moderation with less reliance on manual review for every single upload. **Exam Clue Phrases:** - “Detect objects, faces, or scenes in images or video” - “Content moderation for images” - “Identify what is in a picture” - “Facial analysis or comparison” **Commonly Confused With:** Amazon Textract - Rekognition analyzes visual content (what is in the picture); Textract extracts structured text and data from document images (what is written on the page).

Amazon Textract

Category: Machine Learning **Exam Priority:** Medium **Domain(s):** Domain 1 **What It Is (Plain English):** Amazon Textract automatically extracts printed text, handwriting, and structured data like form fields and table values from scanned documents, images, and PDFs. **Everyday Analogy:** Textract is like a meticulous data-entry clerk who can instantly type out everything written on a scanned form, including the values inside the boxes and tables. **Primary Use Case:** Organizations use Textract to digitize paper forms, invoices, and receipts into structured, searchable data without manual transcription. **Why the AI Practitioner Needs to Know It:** Textract is frequently paired with Comprehend in a two-step pipeline question: Textract extracts the text first, then Comprehend analyzes it. **Business Scenario:** An insurance company received thousands of paper claim forms each month. They used Amazon Textract to automatically extract the form field data into their claims processing system. The result was a dramatic reduction in manual data entry and faster claims turnaround. **Exam Clue Phrases:** - “Extract text from scanned documents or forms” - “Digitize handwritten or printed forms” - “Pull data out of tables in a PDF” - “Automate manual data entry from documents” **Commonly Confused With:** Amazon Comprehend - Textract extracts text from document images into usable text; Comprehend then analyzes the

meaning, sentiment, and entities within text that is already in text form.

Amazon Transcribe

Category: Machine Learning **Exam Priority:** High **Domain(s):** Domain 1 **What It Is (Plain English):** Amazon Transcribe is an automatic speech recognition (ASR) service that converts spoken audio, from calls, recordings, or live streams, into accurate written text transcripts. **Everyday Analogy:** Transcribe is like a court stenographer who listens to a conversation and types out exactly what was said, in real time or after the fact. **Primary Use Case:** Call centers and media companies use Transcribe to generate searchable, written transcripts of phone calls, meetings, or videos. **Why the AI Practitioner Needs to Know It:** Transcribe is the direct opposite of Polly, and this contrast appears repeatedly on the exam as a quick-distinguishing pair. **Business Scenario:** A call center wanted to analyze customer service calls for quality and compliance without listening to every recording manually. They used Amazon Transcribe to convert all call audio into text transcripts automatically. The result was the ability to search, analyze, and audit calls at a scale that would be impossible for a human team alone. **Exam Clue Phrases:** - “Convert spoken audio into text” - “Speech-to-text” - “Transcribe a call or meeting recording” - “Searchable transcripts of audio content” **Commonly Confused With:** Amazon Polly - Transcribe converts speech to text; Polly converts text to speech. They work in opposite directions.

Amazon Translate

Category: Machine Learning **Exam Priority:** High **Domain(s):** Domain 1 **What It Is (Plain English):** Amazon Translate is a neural machine translation service that provides fast, high-quality language translation between dozens of supported languages. **Everyday Analogy:** Translate is like having a professional interpreter instantly available for any written content, in nearly any language pair. **Primary Use Case:** Companies use Translate to localize websites, product listings, or customer communications for international audiences. **Why the AI Practitioner Needs to Know It:** Translate is the standard answer for “convert content from one language to another” and is contrasted with Comprehend, which analyzes meaning within a single language. **Business Scenario:** An e-commerce company wanted to sell products in twelve new countries without hiring translators for every product description. They used Amazon Translate to automatically localize their product catalog into each target language. The result was a faster international launch at a fraction of the translation cost. **Exam Clue Phrases:** - “Translate content into another language” - “Localize content for global audiences” - “Multilingual support for an application” - “Convert text between languages” **Commonly Confused With:** Amazon Comprehend - Translate converts text from one language to another; Comprehend analyzes text for meaning, entities, and sentiment within a single language.

AWS Transform

Category: Machine Learning **Exam Priority:** Medium **Domain(s):** Domain 1 **What It Is (Plain English):** AWS Transform is an AI-powered service that helps organizations modernize and migrate legacy application code, accelerating digital transformation through automated code

analysis and conversion. **Everyday Analogy:** AWS Transform is like an expert translator for old computer languages, automatically converting outdated code into a modern equivalent. **Primary Use Case:** Enterprises with large amounts of legacy application code use AWS Transform to accelerate the analysis and conversion process during a cloud migration or modernization project. **Why the AI Practitioner Needs to Know It:** AWS Transform is a newer addition to the exam's in-scope list and is distinguished from AWS Migration Hub, a broader migration-tracking tool that does not itself use AI to convert code. **Business Scenario:** A bank had decades-old mainframe application code it needed to modernize. They used AWS Transform to accelerate the analysis and conversion of that legacy code using AI. The result was a significantly faster modernization timeline compared to a fully manual rewrite. **Exam Clue Phrases:** - "AI-powered code modernization" - "Convert legacy application code" - "Accelerate digital transformation" - "Automated code analysis and conversion" **Commonly Confused With:** AWS Migration Hub - AWS Transform uses AI to modernize and convert application code; Migration Hub is a tracking and orchestration service for broader cloud migration projects (and is not itself an AI service).

Amazon Q

Category: Developer Tools **Exam Priority:** High **Domain(s):** Domain 1, Domain 2 **What It Is (Plain English):** Amazon Q is an AI-powered assistant from AWS designed to help developers, IT professionals, and business users with coding, troubleshooting, answering questions about AWS, and querying enterprise data. **Everyday Analogy:** Amazon Q is like having a knowledgeable AWS-savvy colleague sitting next to you, ready to answer questions or help you work faster across your tools. **Primary Use Case:** Developers and business users use Amazon Q to accelerate coding, get answers about AWS services, and query internal company data conversationally. **Why the AI Practitioner Needs to Know It:** The exam guide names "Amazon Q" generally as an in-scope service; you should understand its role as an AI assistant rather than memorizing every individual product variation. **Business Scenario:** A software development team wanted to speed up their coding and reduce time spent searching AWS documentation. They used Amazon Q to get inline coding help and instant answers about AWS service configuration. The result was faster development cycles and less time lost to context-switching. **Exam Clue Phrases:** - "AI assistant for developers" - "Ask questions about AWS services" - "AI assistant for business users querying enterprise data" - "Coding help and troubleshooting assistant" **Commonly Confused With:** Amazon Lex - Amazon Q is an AI assistant for employees and developers working within AWS tools and enterprise systems; Lex is a platform for building custom-branded conversational bots for your own end-user-facing applications.

Amazon Q Product Variations (Context Only)

Variation	What It Focuses On	Exam Relevance
Amazon Q Business	Answering questions and automating tasks using a company's enterprise data	The exam guide references "Amazon Q" generally; these named variations are useful context but are not separately itemized on the official in-scope list

Variation	What It Focuses On	Exam Relevance
Amazon Q Developer (formerly associated with Amazon CodeWhisperer branding)	AI-powered coding assistance inside IDEs like VS Code and JetBrains	Same as above; understand the general capability, not brand-specific trivia
Amazon Q in Connect	Real-time assistance for contact center agents during customer interactions	Same as above
Amazon Q in QuickSight (now Amazon Quick)	Natural language querying of business intelligence dashboards	Same as above; also connects to the separately in-scope “Amazon Quick” service

Kiro

Category: Developer Tools **Exam Priority:** Medium **Domain(s):** Domain 1, Domain 2 **What It Is (Plain English):** Kiro is an AI-native software development environment (IDE) from AWS that uses AI agents to help developers design, plan, write, test, and refine code and application architecture. **Everyday Analogy:** Kiro is like a workshop that was built from the ground up around having an AI assistant as a full collaborator, rather than bolting an assistant onto an existing workshop. **Primary Use Case:** Development teams that want an agentic, AI-first coding experience use Kiro as their primary IDE rather than adding an AI plugin to a traditional one. **Why the AI Practitioner Needs to Know It:** Kiro is contrasted with Amazon Q Developer: one is a full AI-native IDE, the other is an assistant added into existing IDEs. **Business Scenario:** A software startup wanted to use agentic AI throughout their entire development process, not just for autocomplete suggestions. They adopted Kiro as their core development environment. The result was a development workflow where AI agents participated in planning and architecture decisions, not just code completion. **Exam Clue Phrases:** - “AI-native development environment” - “AI agents help plan and write code” - “Full IDE built around agentic workflows” **Commonly Confused With:** Amazon Q Developer - Kiro is a full AI-native IDE designed around agentic workflows; Amazon Q Developer is an AI coding assistant that works within existing IDEs like VS Code and JetBrains.

Strands Agents

Category: Developer Tools **Exam Priority:** Medium **Domain(s):** Domain 2, Domain 3 **What It Is (Plain English):** Strands Agents is an open-source AWS framework that provides building blocks for creating AI agents capable of using tools, managing memory, accessing data, and completing multi-step tasks autonomously. **Everyday Analogy:** Strands Agents is like a box of LEGO bricks specifically designed for building robots; you assemble the agent yourself using the provided building blocks. **Primary Use Case:** Developers building custom agentic AI applications use Strands Agents as the open-source foundation for defining how their agent thinks, remembers, and acts. **Why the AI Practitioner Needs to Know It:** The exam pairs Strands Agents (the open-source development framework) against Bedrock AgentCore (the managed production runtime) as a classic “build vs. run” distinction. **Business Scenario:** A software engineering team wanted

full control over how their custom AI agent reasoned and used tools. They used Strands Agents to build the agent’s logic from open-source building blocks. The result was a highly customized agent design, which they then deployed into production using Amazon Bedrock AgentCore. **Exam Clue Phrases:** - “Open-source framework for building agents” - “Building blocks for agent tool usage and memory” - “Developer framework for agentic AI” **Commonly Confused With:** Amazon Bedrock AgentCore - Strands Agents is the open-source development framework for creating agents; AgentCore is the managed AWS service for deploying, running, and securing those agents in production.

SECTION 1B: SUPPLEMENTAL ML/AI SERVICES

The services below are NOT on the official AIF-C01 in-scope services list. They are included for general AWS literacy and context only. Treat exam relevance as minimal to none.

Amazon Forecast

Category: Machine Learning (Supplemental) **Exam Priority:** Not Officially In-Scope **Domain(s):** Not officially mapped (conceptually related to Domain 1’s “regression” and “time-series data” topics) **What It Is (Plain English):** Amazon Forecast was an AWS service that used ML to generate time-series forecasts, such as predicting future demand, sales, or resource needs. **Everyday Analogy:** Forecast is like a weather forecaster, but instead of predicting rain, it predicts business numbers like next month’s product demand. **Primary Use Case:** Businesses used Forecast for demand planning, inventory management, and financial forecasting based on historical time-series data. **Why the AI Practitioner Needs to Know It:** It is not on the official in-scope list, but the underlying concept (time-series forecasting as a use case for regression-style ML) does appear in Domain 1. Know the concept; do not expect this specific service name on the exam. **Business Scenario:** A retailer wanted to avoid both stockouts and overstocking. They used Amazon Forecast to predict product demand using historical sales data. The result was more efficient inventory planning. (This scenario illustrates the underlying concept rather than an exam-tested service.) **Exam Clue Phrases:** - Not applicable for AIF-C01; the concept of “time-series forecasting” itself is testable, but not this specific service name. **Commonly Confused With:** Amazon SageMaker AI’s built-in forecasting algorithms - both can perform time-series prediction, but only SageMaker AI is officially in-scope.

AWS DeepRacer

Category: Machine Learning (Supplemental) **Exam Priority:** Not Officially In-Scope **Domain(s):** Not officially mapped (conceptually related to Domain 1’s “reinforcement learning” topic) **What It Is (Plain English):** AWS DeepRacer is a fully autonomous 1/18th-scale race car designed to help people learn reinforcement learning concepts hands-on. **Everyday Analogy:** DeepRacer is like a video game version of teaching a dog tricks: the car gets a reward signal when it does well and gradually learns better racing behavior through trial and error. **Primary Use Case:** DeepRacer is primarily an educational and competitive tool for learning reinforcement learning concepts, not a production business solution. **Why the AI Practitioner Needs to**

Know It: DeepRacer is not on the in-scope list, but it is a commonly cited example of reinforcement learning in AWS training materials. Know the reinforcement learning concept itself; this service name is unlikely to appear on the exam. **Business Scenario:** A training program wanted a fun, hands-on way to teach reinforcement learning concepts to new engineers. They used AWS DeepRacer’s simulated racing environment as a teaching tool. The result was faster intuitive understanding of reward signals and trial-and-error learning. (Illustrative example only.) **Exam Clue Phrases:** - Not applicable for AIF-C01; the concept of “reinforcement learning” itself is testable, but not this specific service name. **Commonly Confused With:** Reinforcement learning (the general ML technique) - DeepRacer is simply one hands-on educational application of that broader, exam-relevant concept.

Amazon Lookout for Vision

Category: Machine Learning (Supplemental) **Exam Priority:** Not Officially In-Scope **Domain(s):** Not officially mapped (conceptually related to Domain 1’s “computer vision” topic) **What It Is (Plain English):** Amazon Lookout for Vision was a computer vision service that used ML to identify visual defects, such as damaged products or malfunctioning equipment, in images. **Everyday Analogy:** It works like a quality-control inspector on a factory line who can spot a scratch or dent in a product faster than the human eye. **Primary Use Case:** Manufacturers used Lookout for Vision to automatically detect defects on production lines. **Why the AI Practitioner Needs to Know It:** Not on the in-scope list. Amazon Rekognition is the officially in-scope computer vision service you should rely on for exam answers about image analysis. **Business Scenario:** A manufacturer wanted to catch defective products before shipment. They used Amazon Lookout for Vision to automatically flag products with visible defects. The result was fewer defective items reaching customers. (Illustrative example only.) **Exam Clue Phrases:** - Not applicable for AIF-C01. **Commonly Confused With:** Amazon Rekognition - both are computer vision services, but only Rekognition is officially in-scope for this exam.

Amazon Lookout for Metrics

Category: Machine Learning (Supplemental) **Exam Priority:** Not Officially In-Scope **Domain(s):** Not officially mapped **What It Is (Plain English):** Amazon Lookout for Metrics was a service that used ML to automatically detect unusual changes (anomalies) in business and operational metrics. **Everyday Analogy:** It works like a smoke detector for your business data, automatically flagging anything that looks abnormal before it becomes a bigger problem. **Primary Use Case:** Businesses used Lookout for Metrics to catch sudden, unexplained changes in metrics like revenue, traffic, or operational performance. **Why the AI Practitioner Needs to Know It:** Not on the in-scope list. This is included only for general AWS AI/ML literacy. **Business Scenario:** An e-commerce company wanted to catch unusual drops in conversion rate quickly. They used Amazon Lookout for Metrics to flag anomalies automatically. The result was faster detection of issues before they significantly affected revenue. (Illustrative example only.) **Exam Clue Phrases:** - Not applicable for AIF-C01. **Commonly Confused With:** Amazon CloudWatch anomaly detection - both detect unusual patterns, but CloudWatch’s general monitoring capabilities are more relevant to the officially in-scope content in Domain 5.

Amazon Fraud Detector

Category: Machine Learning (Supplemental) **Exam Priority:** Not Officially In-Scope **Domain(s):** Not officially mapped (conceptually related to Domain 1's "fraud detection" use case) **What It Is (Plain English):** Amazon Fraud Detector was a service that used ML to identify potentially fraudulent online activity, such as fake accounts or suspicious payments. **Everyday Analogy:** It functions like a bank teller trained to spot suspicious behavior, but automated and applied to every single online transaction. **Primary Use Case:** Businesses used Fraud Detector to flag suspicious transactions or account activity in real time. **Why the AI Practitioner Needs to Know It:** Not on the in-scope list. However, "fraud detection" as a use case category is explicitly named in Domain 1's task statements, so know the concept even though this specific service is not tested. **Business Scenario:** An online marketplace wanted to reduce losses from fraudulent purchases. They used Amazon Fraud Detector to flag high-risk transactions before they completed. The result was reduced fraud losses with minimal disruption to legitimate customers. (Illustrative example only.) **Exam Clue Phrases:** - Not applicable for AIF-C01; the use case category "fraud detection" is testable as a concept, but not this specific service name. **Commonly Confused With:** Amazon Comprehend - both can play a role in identifying suspicious patterns, but only Comprehend is officially in-scope.

Amazon HealthLake

Category: Machine Learning (Supplemental) **Exam Priority:** Not Officially In-Scope **Domain(s):** Not officially mapped **What It Is (Plain English):** Amazon HealthLake was a HIPAA-eligible service that helped healthcare organizations store, transform, and analyze health data at scale. **Everyday Analogy:** It functions like a specialized filing system built specifically to organize medical records in a standardized, analysis-ready way. **Primary Use Case:** Healthcare organizations used HealthLake to centralize and structure patient data for analytics and ML use cases. **Why the AI Practitioner Needs to Know It:** Not on the in-scope list. Healthcare-specific AWS services are excluded from this foundational exam's scope. **Business Scenario:** A hospital network wanted to unify patient data from multiple systems for analysis. They used Amazon HealthLake to normalize and store that data. The result was a foundation for population health analytics. (Illustrative example only.) **Exam Clue Phrases:** - Not applicable for AIF-C01. **Commonly Confused With:** Amazon S3 - both can store healthcare-related data, but S3 is the officially in-scope general storage answer for AIF-C01.

Amazon CodeGuru

Category: Machine Learning / Developer Tools (Supplemental) **Exam Priority:** Not Officially In-Scope **Domain(s):** Not officially mapped **What It Is (Plain English):** Amazon CodeGuru was a developer tool that used ML to provide automated code reviews and identify the most expensive lines of code in an application. **Everyday Analogy:** It functions like a senior engineer reviewing your pull request, pointing out bugs and performance issues before the code ships. **Primary Use Case:** Development teams used CodeGuru to catch code quality issues and performance bottlenecks automatically during development. **Why the AI Practitioner Needs to Know It:** Not on the in-scope list. Coding and development tasks are explicitly out of scope for the AIF-C01 target candidate, who uses but does not build AI/ML systems. **Business Scenario:** A software

team wanted to catch expensive code patterns before they reached production. They used Amazon CodeGuru to automatically flag costly or risky code during reviews. The result was fewer performance issues discovered after release. (Illustrative example only.) **Exam Clue Phrases:** - Not applicable for AIF-C01. **Commonly Confused With:** Amazon Q Developer - both assist developers, but CodeGuru focuses on automated code review and cost analysis, while Q Developer (under the broader Amazon Q umbrella) focuses on conversational coding assistance.

Amazon CodeWhisperer / Amazon Q Developer

Category: Developer Tools (Supplemental) **Exam Priority:** Not Officially In-Scope (by this specific legacy name) **Domain(s):** Not officially mapped **What It Is (Plain English):** Amazon CodeWhisperer was AWS's original name for its AI coding assistant; its capabilities are now offered under the Amazon Q Developer name, which falls under the broader "Amazon Q" umbrella. **Everyday Analogy:** Think of this as a renamed product, like a restaurant that changed its name but still serves the same menu. **Primary Use Case:** Developers use this tool for AI-assisted code suggestions, completions, and explanations inside their IDE. **Why the AI Practitioner Needs to Know It:** "Amazon Q" (the umbrella service) is officially in-scope. The legacy name "CodeWhisperer" is not separately listed and should not be treated as a distinct exam topic. **Business Scenario:** A development team wanted faster code completions inside their existing editor. They adopted what is now branded Amazon Q Developer for inline coding suggestions. The result was faster development with fewer repetitive typing tasks. (Illustrative example only.) **Exam Clue Phrases:** - Not applicable for AIF-C01 under this name; rely on "Amazon Q" as the officially testable umbrella term. **Commonly Confused With:** Kiro - Q Developer adds AI assistance to an existing IDE; Kiro is a full IDE built natively around agentic AI from the ground up.

SECTION 2: ANALYTICS AND DATA SERVICES

Amazon Athena

Category: Analytics (Supplemental) **Exam Priority:** Not Officially In-Scope **Domain(s):** Not officially mapped **What It Is (Plain English):** Amazon Athena is a serverless query service that lets you analyze data directly in Amazon S3 using standard SQL, without needing to load it into a database first. **Everyday Analogy:** Athena is like being able to ask questions directly to a filing cabinet full of paper, without first having to type everything into a spreadsheet. **Primary Use Case:** Analysts use Athena to run ad hoc SQL queries against data sitting in S3 for quick analysis. **Why the AI Practitioner Needs to Know It:** Not on the official in-scope list. Included here for general AWS data literacy, since it is commonly mentioned alongside other analytics tools used to prepare ML data. **Business Scenario:** A data team wanted to quickly analyze log files stored in S3 without setting up a database. They used Amazon Athena to query the files directly with SQL. The result was faster ad hoc analysis with no infrastructure to manage. (Illustrative example only.) **Exam Clue Phrases:** - Not applicable for AIF-C01. **Commonly Confused With:** Amazon Redshift - Athena queries data directly in S3 without a dedicated warehouse; Redshift is an actual managed data warehouse with its own storage, and is officially in-scope.

AWS Data Exchange

Category: Analytics **Exam Priority:** Low **Domain(s):** Domain 1 **What It Is (Plain English):** AWS Data Exchange is a marketplace where organizations can discover, subscribe to, and use third-party datasets directly within AWS for analytics and ML model training. **Everyday Analogy:** It's like an app store, but instead of apps, you are browsing and subscribing to licensed data. **Primary Use Case:** Organizations that need external data (such as weather, financial, or demographic data) to enrich their own datasets use Data Exchange to subscribe to and access it. **Why the AI Practitioner Needs to Know It:** Know it as the answer when a scenario describes acquiring licensed external data for training or analysis, as opposed to preparing data you already own. **Business Scenario:** A logistics company wanted to factor weather conditions into its delivery time predictions but didn't have its own weather data. They used AWS Data Exchange to subscribe to a third-party weather dataset. The result was more accurate delivery predictions without building a weather-monitoring system themselves. **Exam Clue Phrases:** - "Subscribe to third-party datasets" - "Acquire external/licensed data" - "Data marketplace" **Commonly Confused With:** AWS Glue - Data Exchange is for acquiring licensed external data; Glue is for preparing, transforming, and loading data you already have.

Amazon EMR

Category: Analytics **Exam Priority:** Low **Domain(s):** Domain 1 **What It Is (Plain English):** Amazon EMR is a managed big data platform that runs open-source frameworks like Apache Spark and Hadoop to process very large datasets in parallel. **Everyday Analogy:** EMR is like a massive team of workers processing an enormous pile of paperwork simultaneously, rather than one person working through it alone. **Primary Use Case:** Organizations with very large, complex datasets use EMR for heavy-duty data preparation pipelines that feed into ML training. **Why the AI Practitioner Needs to Know It:** Know that EMR is the more powerful, more complex option for big data processing, compared to the simpler, serverless AWS Glue. **Business Scenario:** A telecom company needed to process petabytes of call detail records for an ML training dataset. They used Amazon EMR to run distributed Spark jobs across a large cluster. The result was data processing completed in hours rather than weeks. **Exam Clue Phrases:** - "Process massive datasets in parallel" - "Apache Spark or Hadoop" - "Large-scale cluster-based data processing" **Commonly Confused With:** AWS Glue - EMR is a full cluster-based big data platform requiring more configuration; Glue is a simpler, serverless ETL service better suited to standard data integration tasks.

AWS Glue

Category: Analytics **Exam Priority:** Medium **Domain(s):** Domain 1 **What It Is (Plain English):** AWS Glue is a serverless data integration service that discovers, prepares, and transforms data from multiple sources, commonly used in the data preparation stage of ML pipelines. **Everyday Analogy:** Glue is like an automatic sorting and labeling machine that takes messy incoming mail and organizes it neatly before it reaches your desk. **Primary Use Case:** Data teams use Glue to clean, transform, and combine data from different sources before it's used for analytics or ML training. **Why the AI Practitioner Needs to Know It:** Glue is the standard answer for "data preparation" or "ETL" (extract, transform, load) questions tied to the AI/ML

pipeline in Domain 1. **Business Scenario:** A healthcare analytics company needed to combine data from several different hospital systems into one clean dataset for ML training. They used AWS Glue to automate the extraction, transformation, and loading process. The result was a consistent, analysis-ready dataset built without manual scripting for every data source. **Exam Clue Phrases:** - “Serverless data integration” - “Prepare and transform data for ML” - “ETL pipeline” - “No cluster to manage” **Commonly Confused With:** Amazon EMR - Glue is simpler and serverless (no cluster to manage); EMR is a more powerful but more complex cluster-based processing platform.

AWS Glue DataBrew

Category: Analytics **Exam Priority:** Low **Domain(s):** Domain 1 **What It Is (Plain English):** AWS Glue DataBrew is a visual, no-code data preparation tool that lets business analysts and data scientists clean and normalize data without writing code. **Everyday Analogy:** DataBrew is like a point-and-click photo editor for data, versus Glue, which is more like writing custom editing scripts. **Primary Use Case:** Teams without strong coding backgrounds use DataBrew to clean and prepare data visually before it's used for analytics or ML. **Why the AI Practitioner Needs to Know It:** Know DataBrew as the “no-code” entry point into data preparation, as distinct from code-based Glue. **Business Scenario:** A marketing analytics team wanted to clean a messy customer dataset but had no engineers available. They used AWS Glue DataBrew's visual interface to fix formatting issues and remove duplicates. The result was a clean dataset prepared without writing a single line of code. **Exam Clue Phrases:** - “No-code data preparation” - “Visual interface for cleaning data” - “Business analysts preparing data without coding” **Commonly Confused With:** AWS Glue - DataBrew is the visual, point-and-click version of data preparation; Glue is the code-based serverless ETL service typically used by engineers.

AWS Lake Formation

Category: Analytics **Exam Priority:** Low **Domain(s):** Domain 1 **What It Is (Plain English):** AWS Lake Formation helps organizations quickly build, secure, and manage a central data lake in Amazon S3, with fine-grained access controls, data cataloging, and governance features. **Everyday Analogy:** Lake Formation is like the security desk and card-catalog system for a giant warehouse (your data lake), controlling exactly who can access which sections. **Primary Use Case:** Organizations centralizing data from many sources into S3 use Lake Formation to apply consistent governance and access controls. **Why the AI Practitioner Needs to Know It:** Know it as the governance layer on top of S3-based data lakes, distinct from S3 itself, which is just storage. **Business Scenario:** A large enterprise had dozens of teams accessing a shared data lake with no consistent permission structure. They used AWS Lake Formation to apply fine-grained access controls and a unified data catalog. The result was tighter security without slowing down legitimate data access. **Exam Clue Phrases:** - “Build and secure a data lake” - “Fine-grained access control over data” - “Centralized data cataloging and governance” **Commonly Confused With:** Amazon S3 - S3 stores the raw data; Lake Formation adds a governance, cataloging, and access control layer on top of that storage.

Amazon OpenSearch Service

Category: Analytics **Exam Priority:** High **Domain(s):** Domain 3 **What It Is (Plain English):** Amazon OpenSearch Service is a managed search and analytics engine that can store and query vector embeddings, making it a common choice for the vector database component of RAG-based applications. **Everyday Analogy:** If embeddings are like fingerprints of meaning, OpenSearch is the fingerprint database that lets you quickly find the closest match. **Primary Use Case:** Organizations building custom RAG applications use OpenSearch to store and search vector embeddings representing their documents. **Why the AI Practitioner Needs to Know It:** OpenSearch is one of four named AWS services for vector storage in RAG (alongside Aurora, Neptune, and RDS for PostgreSQL), and is a high-priority Domain 3 topic. **Business Scenario:** A software company wanted to build a custom RAG application to ground a chatbot in their technical documentation. They used Amazon OpenSearch Service to store and search vector embeddings of that documentation. The result was accurate, grounded chatbot answers pulled directly from their own technical content. **Exam Clue Phrases:** - “Store vector embeddings for RAG” - “Vector database” - “Flexible search and analytics engine” - “Custom RAG implementation” **Commonly Confused With:** Amazon Kendra - OpenSearch is a flexible engine used for vector storage, log analysis, and custom search; Kendra is a higher-level managed intelligent search service purpose-built for enterprise document Q&A.

Amazon Quick

Category: Analytics **Exam Priority:** Medium **Domain(s):** Domain 1, Domain 2 **What It Is (Plain English):** Amazon Quick is a business intelligence and data visualization service that enables interactive dashboards and reports, with AI-powered natural language query capabilities for non-technical users. **Everyday Analogy:** Quick is like a smart dashboard you can simply talk to, asking “what were our top sales last quarter?” instead of building a chart yourself. **Primary Use Case:** Business teams use Quick to build dashboards and let non-technical users ask plain-language questions about their data. **Why the AI Practitioner Needs to Know It:** Know it as the BI and visualization layer, distinct from the underlying data warehouse (Redshift) where the actual data lives. **Business Scenario:** A sales organization wanted managers to be able to ask data questions without waiting on the analytics team. They used Amazon Quick’s natural language query feature to let managers ask questions directly. The result was faster access to insights without bottlenecking through a centralized reporting team. **Exam Clue Phrases:** - “Business intelligence dashboards” - “Natural language queries on business data” - “Visualize data for non-technical users” **Commonly Confused With:** Amazon Redshift - Quick is the front-end BI and visualization layer; Redshift is the cloud data warehouse where the underlying analytical data is actually stored and queried.

Amazon Redshift

Category: Analytics **Exam Priority:** Low **Domain(s):** Domain 1 **What It Is (Plain English):** Amazon Redshift is a fully managed cloud data warehouse optimized for analytical queries across large volumes of structured data. **Everyday Analogy:** Redshift is like a massive, organized records archive built specifically for answering complex historical questions quickly, not for everyday transaction processing. **Primary Use Case:** Organizations use Redshift to store and

query large volumes of historical business data for reporting and analytics, including data later used in ML training. **Why the AI Practitioner Needs to Know It:** Know Redshift as an analytical data warehouse, contrasted with RDS, which handles day-to-day transactional workloads. **Business Scenario:** A retail chain needed to analyze years of sales history to train a demand forecasting model. They used Amazon Redshift to store and query that historical data efficiently. The result was a reliable data source for both business reporting and ML model training. **Exam Clue Phrases:** - “Data warehouse for analytics” - “Query large volumes of historical data” - “Source of ML training data at scale” **Commonly Confused With:** Amazon RDS - Redshift is an analytical data warehouse designed for complex queries over large historical datasets; RDS is an operational relational database designed for transactional workloads.

SECTION 3: SECURITY, IDENTITY, AND COMPLIANCE SERVICES

AWS Identity and Access Management (IAM)

Category: Security, Identity, and Compliance **Exam Priority:** High **Domain(s):** Domain 4, Domain 5 **What It Is (Plain English):** AWS IAM is the foundational AWS service for controlling who, users, roles, and services, can access which AWS resources and what actions they can perform. **Everyday Analogy:** IAM is like a building’s keycard system: it decides who gets through which doors and what they’re allowed to do once inside. **Primary Use Case:** Every AWS account uses IAM to define permissions, ensuring people and systems can only access what they need. **Why the AI Practitioner Needs to Know It:** IAM is foundational to AI security in Domain 5: questions about restricting access to models, training data, or Bedrock agents almost always point back to IAM roles and policies. **Business Scenario:** A company building a GenAI application wanted to ensure only the application’s backend service, not individual employees, could call their foundation model directly. They used AWS IAM to create a tightly scoped role with only the necessary permissions. The result was a properly secured application with no unnecessary access to sensitive AI resources. **Exam Clue Phrases:** - “Control who can access AI resources” - “Least privilege access” - “Roles, policies, and permissions” - “Restrict access to a model or dataset” **Commonly Confused With:** AWS Secrets Manager - IAM controls access to AWS services and resources; Secrets Manager securely stores and retrieves application credentials like database passwords and API keys.

Amazon Macie

Category: Security, Identity, and Compliance **Exam Priority:** High **Domain(s):** Domain 4, Domain 5 **What It Is (Plain English):** Amazon Macie is an ML-powered data security service that automatically discovers and classifies sensitive data, such as PII, financial records, and health information, stored in Amazon S3. **Everyday Analogy:** Macie is like a security scanner that walks through a warehouse of boxes and flags exactly which ones contain sensitive documents, without you having to open each box yourself. **Primary Use Case:** Organizations use Macie to find and protect sensitive data stored in S3 before it’s used in AI training pipelines or accidentally exposed. **Why the AI Practitioner Needs to Know It:** Macie is a key Domain 5 answer for “discover sensitive data in storage,” and is frequently paired against Inspector (which scans

compute, not storage). **Business Scenario:** A healthcare technology company wanted to confirm that no unprotected patient data had accidentally ended up in a general-purpose S3 bucket used for ML experimentation. They used Amazon Macie to scan and classify the contents of that bucket. The result was early detection of sensitive data, allowing the company to fix the issue before it became a compliance incident. **Exam Clue Phrases:** - “Discover sensitive data in S3” - “Detect PII automatically” - “Classify sensitive data before training” - “Data leakage prevention in storage” **Commonly Confused With:** Amazon Inspector - Macie focuses on discovering sensitive data stored in S3 buckets; Inspector focuses on scanning compute resources for software vulnerabilities.

AWS Key Management Service (AWS KMS)

Category: Security, Identity, and Compliance **Exam Priority:** Medium **Domain(s):** Domain 5 **What It Is (Plain English):** AWS KMS is a managed service for creating, storing, and controlling the encryption keys used to protect data at rest and in transit. **Everyday Analogy:** KMS is like a highly secure key-cutting and key-storage service: it manages the actual keys that lock and unlock your data. **Primary Use Case:** Organizations use KMS to encrypt sensitive data, including training datasets and model artifacts, both when stored and when moving across networks. **Why the AI Practitioner Needs to Know It:** KMS is the standard exam answer for “manage encryption keys,” distinct from Secrets Manager, which manages application credentials rather than the keys themselves. **Business Scenario:** A financial services firm needed to encrypt sensitive training data used to build a credit-risk model. They used AWS KMS to manage the encryption keys protecting that data both at rest and in transit. The result was compliance with strict data protection requirements without building custom encryption infrastructure. **Exam Clue Phrases:** - “Manage encryption keys” - “Encrypt data at rest and in transit” - “Control who can decrypt sensitive data” **Commonly Confused With:** AWS Secrets Manager - KMS manages the cryptographic keys used to encrypt and decrypt data; Secrets Manager manages application secrets like passwords and API tokens that applications need to authenticate.

AWS Secrets Manager

Category: Security, Identity, and Compliance **Exam Priority:** Low **Domain(s):** Domain 5 **What It Is (Plain English):** AWS Secrets Manager securely stores, manages, retrieves, and automatically rotates application secrets such as database credentials, API keys, and OAuth tokens. **Everyday Analogy:** Secrets Manager is like a digital safe deposit box for your application’s passwords and keys, with the locks automatically changed on a schedule. **Primary Use Case:** Applications use Secrets Manager to securely retrieve credentials at runtime instead of hardcoding them into application code. **Why the AI Practitioner Needs to Know It:** Know it as the place application credentials live, distinct from KMS, which manages the encryption keys themselves. **Business Scenario:** A development team building a GenAI application needed to securely store the API key used to call a third-party data service. They used AWS Secrets Manager to store and automatically rotate that credential. The result was reduced risk of credential leakage compared to storing the key directly in code. **Exam Clue Phrases:** - “Securely store application credentials” - “Automatic credential rotation” - “API keys and database passwords” **Commonly Confused With:** AWS KMS - Secrets Manager stores application credentials and secrets that code needs to authenticate to services; KMS manages the cryptographic keys used to encrypt and decrypt data

generally.

Amazon Inspector

Category: Security, Identity, and Compliance **Exam Priority:** Medium **Domain(s):** Domain 5 **What It Is (Plain English):** Amazon Inspector is an automated vulnerability management service that continuously scans AWS compute workloads, like EC2 instances and containers, for known software vulnerabilities and unintended network exposure. **Everyday Analogy:** Inspector is like a building inspector who regularly checks the structural integrity of your compute “buildings,” looking for cracks before they become collapses. **Primary Use Case:** Organizations running self-hosted AI workloads on EC2 or in containers use Inspector to find and fix vulnerabilities before they’re exploited. **Why the AI Practitioner Needs to Know It:** Inspector and Macie are a frequent exam pairing: one scans compute, the other scans storage for sensitive data. **Business Scenario:** A company self-hosting an open-source foundation model on EC2 instances wanted to ensure those instances had no known security vulnerabilities. They used Amazon Inspector to continuously scan the instances. The result was earlier detection of vulnerabilities, reducing the window of exposure to potential attacks. **Exam Clue Phrases:** - “Scan compute workloads for vulnerabilities” - “Detect unintended network exposure” - “Vulnerability management for EC2 or containers” **Commonly Confused With:** Amazon Macie - Inspector scans compute resources for security vulnerabilities; Macie scans Amazon S3 storage for sensitive data that should not be exposed.

AWS Artifact

Category: Security, Identity, and Compliance **Exam Priority:** High **Domain(s):** Domain 5 **What It Is (Plain English):** AWS Artifact is a self-service portal where customers can access and download AWS’s own compliance reports, security certifications, and audit documentation, at no cost. **Everyday Analogy:** Artifact is like a public records office where you can download an official report verifying that a building meets fire and safety codes, except the “building” is AWS’s own infrastructure. **Primary Use Case:** Compliance teams use Artifact to retrieve evidence of AWS’s own security certifications to satisfy auditors or regulators. **Why the AI Practitioner Needs to Know It:** Artifact is consistently contrasted with Audit Manager: Artifact gives you AWS’s reports about AWS; Audit Manager helps you build evidence about your own environment. **Business Scenario:** A healthcare company deploying an AI application needed to prove to an auditor that AWS’s underlying infrastructure met specific security certifications. They used AWS Artifact to download the relevant compliance reports. The result was a faster audit process, since they didn’t need to independently verify AWS’s own infrastructure controls. **Exam Clue Phrases:** - “Download AWS compliance reports” - “Access AWS security certifications” - “Self-service portal for audit documentation” - “No-cost access to AWS attestation reports” **Commonly Confused With:** AWS Audit Manager - Artifact is where customers download AWS’s own pre-existing compliance reports and certifications; Audit Manager helps customers build and maintain their own internal compliance evidence.

AWS Audit Manager

Category: Security, Identity, and Compliance **Exam Priority:** High **Domain(s):** Domain 5
What It Is (Plain English): AWS Audit Manager continuously and automatically collects evidence of AWS resource configurations and activity, helping organizations demonstrate compliance with frameworks like SOC 2, GDPR, and HIPAA. **Everyday Analogy:** Audit Manager is like an assistant who is constantly photographing and filing evidence of your own business practices, so you have a ready-made folder when an inspector shows up. **Primary Use Case:** Organizations subject to ongoing compliance requirements use Audit Manager to continuously gather evidence about their own environment, rather than scrambling before each audit. **Why the AI Practitioner Needs to Know It:** Know the direction of responsibility: Audit Manager documents what the customer's environment is doing, not what AWS's infrastructure is doing. **Business Scenario:** A fintech company needed to continuously prove compliance with SOC 2 requirements for its AI-powered fraud detection system. They used AWS Audit Manager to automatically collect evidence of their resource configurations over time. The result was a continuously audit-ready environment instead of a stressful, last-minute compliance scramble. **Exam Clue Phrases:** - "Continuously collect compliance evidence" - "Demonstrate compliance with SOC 2, GDPR, or HIPAA" - "Automated evidence collection for your own environment" **Commonly Confused With:** AWS Artifact - Audit Manager is used by customers to conduct and document their own ongoing compliance audits; Artifact provides pre-existing AWS audit reports for customers to download.

Amazon GuardDuty

Category: Security, Identity, and Compliance (Supplemental) **Exam Priority:** Not Officially In-Scope **Domain(s):** Not officially mapped **What It Is (Plain English):** Amazon GuardDuty is a threat detection service that continuously monitors AWS accounts and workloads for malicious activity and unauthorized behavior. **Everyday Analogy:** GuardDuty is like a 24/7 security monitoring service that watches for suspicious activity across your entire property, not just one building. **Primary Use Case:** Security teams use GuardDuty to detect compromised credentials, unusual API calls, or other signs of an active attack. **Why the AI Practitioner Needs to Know It:** Not on the official in-scope list for AIF-C01. General AWS threat detection is covered conceptually in Domain 5 (threat detection is named explicitly in Task Statement 5.1), but GuardDuty itself is not named as a tested service. **Business Scenario:** A company running AI workloads in the cloud wanted continuous monitoring for compromised credentials or unusual account activity. They used Amazon GuardDuty to detect such threats automatically. The result was faster detection of potential security incidents. (Illustrative example only.) **Exam Clue Phrases:** - Not applicable for AIF-C01 by this specific service name; the broader concept of "threat detection" is testable. **Commonly Confused With:** Amazon Inspector - GuardDuty detects active threats and suspicious behavior; Inspector proactively scans for vulnerabilities before they're exploited. Inspector is officially in-scope; GuardDuty is not.

SECTION 4: MANAGEMENT AND GOVERNANCE SERVICES

AWS CloudTrail

Category: Management and Governance **Exam Priority:** High **Domain(s):** Domain 5 **What It Is (Plain English):** AWS CloudTrail records a complete history of all API calls and actions taken within an AWS account by users, roles, and services, creating an auditable activity log. **Everyday Analogy:** CloudTrail is like a security camera log that records every single action taken in a building: who entered which room, when, and what they did there. **Primary Use Case:** Organizations use CloudTrail to investigate security incidents, support audits, and maintain accountability for who did what within their AWS environment. **Why the AI Practitioner Needs to Know It:** CloudTrail is consistently contrasted with CloudWatch: CloudTrail answers “who did what,” while CloudWatch answers “how is the system performing.” **Business Scenario:** A company needed to investigate who had modified the configuration of their Bedrock Guardrails after an unexpected change in model behavior. They used AWS CloudTrail to review the full history of API calls in their account. The result was a clear audit trail identifying exactly who made the change and when. **Exam Clue Phrases:** - “Audit trail of account activity” - “Who made this API call?” - “Log of actions taken in the AWS account” - “Investigate a security incident” **Commonly Confused With:** Amazon CloudWatch - CloudTrail logs who did what and when (API and account activity for auditing); CloudWatch monitors operational performance metrics, resource health, and application logs.

Amazon CloudWatch

Category: Management and Governance **Exam Priority:** Medium **Domain(s):** Domain 4, Domain 5 **What It Is (Plain English):** Amazon CloudWatch is a monitoring and observability service that collects metrics, logs, and events from AWS resources and applications, enabling automated alerts and dashboards. **Everyday Analogy:** CloudWatch is like the dashboard gauges in a car: it constantly shows you speed, fuel level, and engine temperature so you know when something needs attention. **Primary Use Case:** Teams use CloudWatch to monitor the operational health and performance of deployed AI applications and infrastructure in real time. **Why the AI Practitioner Needs to Know It:** CloudWatch represents ongoing operational monitoring, which connects to both MLOps concepts (Domain 1) and model/system health monitoring (Domains 4 and 5). **Business Scenario:** A company running a production GenAI application wanted to know immediately if response latency spiked or error rates increased. They used Amazon CloudWatch to set up dashboards and automated alerts. The result was faster detection and response to operational issues before customers were significantly affected. **Exam Clue Phrases:** - “Monitor operational performance” - “Set up alerts and dashboards” - “Track resource health metrics” **Commonly Confused With:** AWS CloudTrail - CloudWatch monitors what is happening operationally (performance and health); CloudTrail records what actions were taken by whom (activity and change history).

AWS Config

Category: Management and Governance **Exam Priority:** High **Domain(s):** Domain 5 **What It Is (Plain English):** AWS Config continuously records and tracks the configuration state of

AWS resources over time, enabling compliance evaluation against rules and identification of when and how configurations changed. **Everyday Analogy:** Config is like a detailed change log for a building's floor plan: it tracks every time a wall is moved or a door is added, and can flag changes that violate code. **Primary Use Case:** Compliance and security teams use Config to ensure AI infrastructure stays configured according to organizational policy, and to detect drift when it doesn't. **Why the AI Practitioner Needs to Know It:** Config is frequently contrasted with CloudTrail: Config tracks the resulting state of a resource, while CloudTrail tracks the specific action that caused the change. **Business Scenario:** A company wanted to ensure that no S3 bucket containing training data was ever accidentally made publicly accessible. They used AWS Config to continuously evaluate bucket configurations against a compliance rule. The result was automatic detection of any non-compliant configuration the moment it occurred. **Exam Clue Phrases:** - "Track resource configuration over time" - "Evaluate compliance against rules" - "Detect configuration drift" **Commonly Confused With:** AWS CloudTrail - Config tracks the state of resource configurations and whether they comply with rules; CloudTrail tracks the API actions that caused those configuration changes to happen.

AWS Trusted Advisor

Category: Management and Governance **Exam Priority:** Medium **Domain(s):** Domain 5 **What It Is (Plain English):** AWS Trusted Advisor is an automated tool that analyzes an AWS environment across five pillars and provides real-time, actionable recommendations for cost, security, fault tolerance, performance, and service limits. **Everyday Analogy:** Trusted Advisor is like a financial advisor who automatically reviews your accounts and flags simple ways you could save money or reduce risk, without you having to ask. **Primary Use Case:** Teams use Trusted Advisor for ongoing, automated, no-effort recommendations to improve their AWS environment. **Why the AI Practitioner Needs to Know It:** Know the contrast with the Well-Architected Tool: Trusted Advisor is automated and continuous, while Well-Architected Tool is a structured, human-led review. **Business Scenario:** A growing AI startup wanted ongoing visibility into cost and security improvements without dedicating staff to manually review everything. They used AWS Trusted Advisor to receive continuous, automated recommendations. The result was steady incremental improvements with minimal manual effort. **Exam Clue Phrases:** - "Automated, real-time recommendations" - "Cost savings and security improvement suggestions" - "Continuous checks without a formal review process" **Commonly Confused With:** AWS Well-Architected Tool - Trusted Advisor provides continuous automated checks with immediate recommendations; Well-Architected Tool supports structured, human-led architecture reviews against AWS best practices.

AWS Well-Architected Tool

Category: Management and Governance **Exam Priority:** Low **Domain(s):** Domain 5 **What It Is (Plain English):** The AWS Well-Architected Tool is a guided review tool that helps organizations evaluate their workloads against AWS best practices across six pillars, including security, reliability, and cost optimization. **Everyday Analogy:** It's like a formal building inspection checklist that an architect walks through with you, rather than an automatic alarm system. **Primary Use Case:** Architecture teams use the Well-Architected Tool during structured, periodic

reviews of a workload’s design. **Why the AI Practitioner Needs to Know It:** Know it as the structured, question-driven counterpart to the always-on, automated Trusted Advisor. **Business Scenario:** An organization wanted a formal, structured evaluation of their new AI application’s architecture before scaling it further. They used the AWS Well-Architected Tool to walk through a guided review across all six pillars. The result was a documented set of architectural improvements before the application reached a larger customer base. **Exam Clue Phrases:** - “Structured architecture review” - “Six pillars of the Well-Architected Framework” - “Guided, question-driven workload evaluation” **Commonly Confused With:** AWS Trusted Advisor - Well-Architected Tool is a structured, question-driven review process conducted by architects; Trusted Advisor provides automated, continuous checks without requiring a formal review process.

AWS Control Tower

Category: Management and Governance (Supplemental) **Exam Priority:** Not Officially In-Scope **Domain(s):** Not officially mapped **What It Is (Plain English):** AWS Control Tower automates the setup and governance of a secure, multi-account AWS environment based on best-practice blueprints. **Everyday Analogy:** Control Tower is like a property management company that sets consistent rules and structure across an entire apartment complex, rather than managing one unit at a time. **Primary Use Case:** Large organizations with many AWS accounts use Control Tower to enforce consistent governance and security guardrails across all of them. **Why the AI Practitioner Needs to Know It:** Not on the official in-scope list for AIF-C01. Included here for general AWS governance literacy only. **Business Scenario:** A large enterprise running many separate AI projects across different teams wanted consistent governance guardrails across all of their AWS accounts. They used AWS Control Tower to enforce those guardrails centrally. The result was consistent security posture across every team’s environment. (Illustrative example only.) **Exam Clue Phrases:** - Not applicable for AIF-C01. **Commonly Confused With:** AWS Organizations - both relate to multi-account management, and neither is officially in-scope for AIF-C01; Organizations is the underlying account structure, while Control Tower automates governance on top of it.

AWS Organizations

Category: Management and Governance (Supplemental) **Exam Priority:** Not Officially In-Scope **Domain(s):** Not officially mapped **What It Is (Plain English):** AWS Organizations lets you centrally manage and govern multiple AWS accounts as a single organization, including consolidated billing and account-level policies. **Everyday Analogy:** Organizations is like a parent company’s org chart, grouping subsidiary accounts together for unified management and billing. **Primary Use Case:** Enterprises with multiple AWS accounts use Organizations to manage billing and apply policies centrally. **Why the AI Practitioner Needs to Know It:** Not on the official in-scope list. Included for general AWS account-structure literacy only. **Business Scenario:** A company with separate AWS accounts for development, testing, and production wanted consolidated billing and centralized policy control. They used AWS Organizations to manage all accounts under one structure. The result was simplified billing and consistent policy enforcement. (Illustrative example only.) **Exam Clue Phrases:** - Not applicable for AIF-C01. **Commonly Confused With:** AWS Control Tower - Organizations provides the underlying multi-account structure and policies; Control

Tower automates governance and best-practice setup on top of that structure.

SECTION 5: DEVELOPER TOOLS AND DEPLOYMENT SERVICES

AWS Lambda

Category: Compute **Exam Priority:** Low **Domain(s):** Domain 1, Domain 3 **What It Is (Plain English):** AWS Lambda is a serverless compute service that runs code in response to events without requiring server management, commonly used for lightweight AI inference triggers and glue logic in ML pipelines. **Everyday Analogy:** Lambda is like a vending machine: it sits idle until someone presses a button (an event), does its one job, and then goes back to waiting, with no staff needed to run it. **Primary Use Case:** Developers use Lambda to run small pieces of code automatically in response to events, such as triggering an inference call when a file is uploaded to S3. **Why the AI Practitioner Needs to Know It:** Lambda is the standard exam answer for “serverless” or “event-driven” inferencing scenarios, contrasted against EC2 for sustained workloads. **Business Scenario:** A document processing company wanted to automatically run text extraction the moment a new file was uploaded to storage, without keeping a server running at all times. They used AWS Lambda to trigger that processing only when needed. The result was lower costs since they paid only for the brief moments code actually ran. **Exam Clue Phrases:** - “Serverless, event-driven processing” - “Run code automatically when a file is uploaded” - “No servers to manage” - “Lightweight inference trigger” **Commonly Confused With:** Amazon EC2 - Lambda is event-driven and best suited for short, bursty functions; EC2 provides always-available virtual machines for sustained, long-running compute work.

Amazon ECR

Category: Developer Tools / Containers (Supplemental) **Exam Priority:** Not Officially In-Scope **Domain(s):** Not officially mapped **What It Is (Plain English):** Amazon Elastic Container Registry (ECR) is a managed service for storing, managing, and deploying container images. **Everyday Analogy:** ECR is like a labeled storage locker for shipping containers (Docker images), making it easy to grab the exact version you need when deploying. **Primary Use Case:** Teams running containerized ML models or applications use ECR to store and version the container images used for deployment. **Why the AI Practitioner Needs to Know It:** Not on the official in-scope list for AIF-C01. Containerized model deployment details fall outside this foundational, non-developer exam’s scope. **Business Scenario:** A team deploying a self-hosted model in containers needed a reliable place to store and version their container images. They used Amazon ECR to manage those images. The result was consistent, version-controlled deployments. (Illustrative example only.) **Exam Clue Phrases:** - Not applicable for AIF-C01. **Commonly Confused With:** Amazon ECS / Amazon EKS - ECR stores the container images; ECS and EKS actually run those containers. ECS and EKS are officially in-scope; ECR is not separately named.

AWS Step Functions

Category: Developer Tools (Supplemental) **Exam Priority:** Not Officially In-Scope **Domain(s):** Not officially mapped **What It Is (Plain English):** AWS Step Functions is a serverless orchestration service that lets you coordinate multiple AWS services into a visual, multi-step workflow. **Everyday Analogy:** Step Functions is like a flowchart that actually runs itself, automatically moving a task from one step to the next based on the outcome of each step. **Primary Use Case:** Developers use Step Functions to orchestrate multi-step processes, such as a data processing pipeline or a multi-stage ML workflow. **Why the AI Practitioner Needs to Know It:** Not on the official in-scope list. The general concept of “workflow orchestration” is named in Domain 2’s agentic AI content, but this specific service is not. **Business Scenario:** A company wanted to coordinate a multi-step ML pipeline involving data preparation, model invocation, and result storage. They used AWS Step Functions to orchestrate that workflow visually. The result was a more maintainable and easier-to-debug pipeline. (Illustrative example only.) **Exam Clue Phrases:** - Not applicable for AIF-C01 by this service name; the concept of “workflow orchestration” itself is testable in the context of agentic AI. **Commonly Confused With:** SageMaker Pipelines - both orchestrate multi-step workflows, but SageMaker Pipelines is specifically built for ML pipeline steps within SageMaker AI.

Amazon API Gateway

Category: Developer Tools (Supplemental) **Exam Priority:** Not Officially In-Scope **Domain(s):** Not officially mapped **What It Is (Plain English):** Amazon API Gateway is a fully managed service for creating, publishing, and securing APIs at any scale, often used as the front door for applications calling backend services. **Everyday Analogy:** API Gateway is like the reception desk of an office building: every visitor (request) checks in there before being routed to the right department. **Primary Use Case:** Developers use API Gateway to expose a secure, managed API endpoint for applications, including those that call AI models behind the scenes. **Why the AI Practitioner Needs to Know It:** Not on the official in-scope list for AIF-C01. Building and exposing APIs is a developer-level task outside this exam’s non-developer scope. **Business Scenario:** A company wanted to expose their custom AI inference endpoint securely to a mobile app. They used Amazon API Gateway to manage authentication and request routing to that endpoint. The result was a secure, scalable API layer in front of their AI service. (Illustrative example only.) **Exam Clue Phrases:** - Not applicable for AIF-C01. **Commonly Confused With:** Amazon Bedrock’s own API - Bedrock already exposes a managed API for foundation models; API Gateway would typically be used in front of a custom, self-hosted model instead.

SECTION 6: DATABASE SERVICES

Amazon Aurora

Category: Database **Exam Priority:** High **Domain(s):** Domain 3 **What It Is (Plain English):** Amazon Aurora is a MySQL and PostgreSQL-compatible relational database built for the cloud, with support for the pgvector extension that enables vector embedding storage for RAG applications. **Everyday Analogy:** Aurora is like a high-performance version of a familiar filing system; it works the way you already expect a relational database to work, just faster and with

cloud-native reliability built in. **Primary Use Case:** Organizations needing a high-performance relational database, with the option to also store vector embeddings for RAG, use Aurora. **Why the AI Practitioner Needs to Know It:** Aurora is one of the four named AWS services for storing RAG vector embeddings, and it's the highest-priority database service on this exam. **Business Scenario:** An enterprise software company wanted a single, high-performance database that could handle both their normal transactional data and the vector embeddings for a new RAG feature. They used Amazon Aurora with the pgvector extension to support both needs in one system. The result was simplified infrastructure without standing up a separate dedicated vector database. **Exam Clue Phrases:** - "High-performance relational database for AWS" - "pgvector support for RAG" - "Vector embeddings in a relational database" **Commonly Confused With:** Amazon RDS - Aurora is a high-performance AWS-built database engine hosted within RDS; RDS is the broader managed relational database service that hosts Aurora along with MySQL, PostgreSQL, Oracle, and SQL Server.

Amazon RDS

Category: Database **Exam Priority:** Medium **Domain(s):** Domain 3 **What It Is (Plain English):** Amazon RDS is a managed relational database service supporting multiple engines (MySQL, PostgreSQL, Oracle, SQL Server, MariaDB), with PostgreSQL's pgvector extension also enabling vector storage for AI applications. **Everyday Analogy:** RDS is like a managed apartment building where you can choose from several floor plans (database engines), and the building staff (AWS) handles all the maintenance. **Primary Use Case:** Organizations needing a traditional managed relational database, potentially with vector storage capability via PostgreSQL, use RDS. **Why the AI Practitioner Needs to Know It:** RDS for PostgreSQL with pgvector is explicitly named as a vector storage option for RAG in Domain 3's task statements. **Business Scenario:** A company already running their core application on RDS for PostgreSQL wanted to add a RAG feature without introducing a brand-new database system. They used the pgvector extension within their existing RDS instance to store embeddings. The result was a RAG capability added with minimal new infrastructure. **Exam Clue Phrases:** - "Managed relational database, multiple engine options" - "PostgreSQL with pgvector" - "Traditional transactional database needs" **Commonly Confused With:** Amazon Aurora - RDS is the broader managed relational database service that includes many engine options; Aurora is AWS's own high-performance engine option available within RDS.

Amazon DynamoDB

Category: Database **Exam Priority:** Low **Domain(s):** Domain 3 **What It Is (Plain English):** Amazon DynamoDB is a fully managed, serverless NoSQL key-value and document database designed for single-digit millisecond response times at any scale. **Everyday Analogy:** DynamoDB is like a lightning-fast coat check counter: you hand over a ticket (key), and you get your item back instantly, with no complex lookup required. **Primary Use Case:** Applications requiring extremely fast, simple lookups at massive scale, such as session data or user profile data for an AI application, use DynamoDB. **Why the AI Practitioner Needs to Know It:** Know DynamoDB as the fast, simple-access-pattern option, contrasted with RDS's complex query capabilities. **Business Scenario:** A GenAI chatbot application needed to instantly retrieve a

user's conversation history by user ID for every single request. They used Amazon DynamoDB to store and retrieve that session data with minimal latency. The result was a responsive chatbot experience even under heavy traffic. **Exam Clue Phrases:** - "Single-digit millisecond lookups" - "Serverless NoSQL key-value store" - "Fast, simple access pattern at scale" **Commonly Confused With:** Amazon RDS - DynamoDB is a NoSQL database optimized for speed and scale with simple access patterns; RDS is a relational database optimized for complex queries and transactional integrity.

Amazon DocumentDB

Category: Database **Exam Priority:** Low **Domain(s):** Domain 3 **What It Is (Plain English):** Amazon DocumentDB is a fully managed document database compatible with MongoDB that stores flexible JSON-like documents. **Everyday Analogy:** DocumentDB is like a filing cabinet where each folder (document) can have a different shape and contents, unlike a rigid spreadsheet with fixed columns. **Primary Use Case:** Applications with varying or evolving data structures, such as user profiles or product catalogs, use DocumentDB. **Why the AI Practitioner Needs to Know It:** Know it as the MongoDB-compatible option, distinguished from DynamoDB's simpler key-value model. **Business Scenario:** A company building a product catalog with widely varying attributes per product category needed a flexible data model. They used Amazon DocumentDB to store those varied product documents. The result was a data structure that could evolve without requiring a rigid schema redesign. **Exam Clue Phrases:** - "MongoDB-compatible document database" - "Flexible JSON-like document storage" - "Varying data structures" **Commonly Confused With:** Amazon DynamoDB - DocumentDB uses a MongoDB-compatible document model suited to complex document structures; DynamoDB is a key-value and simple document NoSQL database optimized for single-digit millisecond lookups.

Amazon ElastiCache

Category: Database **Exam Priority:** Low **Domain(s):** Domain 3 **What It Is (Plain English):** Amazon ElastiCache is a fully managed in-memory caching service (Redis or Memcached compatible) used to store frequently accessed data in memory for ultra-low latency retrieval. **Everyday Analogy:** ElastiCache is like keeping your most-used files on your desk instead of in a filing cabinet down the hall, so you can grab them instantly. **Primary Use Case:** Applications use ElastiCache to cache frequently requested data, such as repeated AI prompt responses, to reduce latency and cost. **Why the AI Practitioner Needs to Know It:** Know it as a temporary, in-memory caching layer, distinct from a durable database like DynamoDB. **Business Scenario:** A GenAI application was repeatedly receiving the same common questions, causing unnecessary repeated calls to the foundation model. They used Amazon ElastiCache to cache common responses for ultra-fast retrieval. The result was lower latency and reduced token costs for frequently asked questions. **Exam Clue Phrases:** - "In-memory caching" - "Ultra-low latency data retrieval" - "Cache frequently accessed data" **Commonly Confused With:** Amazon DynamoDB - ElastiCache provides temporary in-memory caching of hot data for microsecond access; DynamoDB is a durable, persistent database for storing records with millisecond access.

Amazon Neptune

Category: Database **Exam Priority:** Medium **Domain(s):** Domain 3 **What It Is (Plain English):** Amazon Neptune is a fully managed graph database service optimized for highly connected data, like social networks or knowledge graphs, and it also supports vector search for AI applications. **Everyday Analogy:** Neptune is like a database that thinks in terms of relationships, “who knows whom” or “what connects to what,” rather than rows and columns. **Primary Use Case:** Applications involving complex relationships between entities, such as fraud detection networks or recommendation graphs, use Neptune. **Why the AI Practitioner Needs to Know It:** Neptune is one of the four named AWS vector storage options for RAG, and it stands out because of its graph-based, relationship-centric model. **Business Scenario:** A company building a knowledge-graph-based RAG application wanted to capture not just documents but the relationships between concepts in those documents. They used Amazon Neptune to store both the graph relationships and supporting vector data. The result was richer, more contextually connected AI responses than a simple document lookup could provide. **Exam Clue Phrases:** - “Graph database for connected data” - “Knowledge graphs” - “Highly connected or relationship-based data” - “Vector search combined with graph relationships” **Commonly Confused With:** Amazon DocumentDB - Neptune is a graph database designed for traversing complex relationships between entities; DocumentDB is a document database for storing and querying flexible document structures.

SECTION 7: COMPUTE SERVICES

Amazon EC2

Category: Compute **Exam Priority:** Low **Domain(s):** Domain 1, Domain 3 **What It Is (Plain English):** Amazon EC2 is the foundational AWS virtual machine service, used to run AI/ML workloads that need dedicated compute, including GPU-based instances for model training and self-hosted model inference. **Everyday Analogy:** EC2 is like renting a fully equipped workshop for as long as you need it, rather than borrowing a tool for a single quick task. **Primary Use Case:** Organizations that need sustained, dedicated compute power, such as training a custom model or self-hosting an open-source model, use EC2. **Why the AI Practitioner Needs to Know It:** EC2 is the standard contrast point against Lambda: sustained workloads versus short, event-driven tasks. **Business Scenario:** A research team needed sustained GPU compute power over several days to train a custom deep learning model. They used Amazon EC2 GPU instances to run that extended training job. The result was a successfully trained model without owning or maintaining physical GPU hardware. **Exam Clue Phrases:** - “Dedicated virtual machine for training” - “GPU instances for model training” - “Self-hosted model inference” - “Sustained compute workload” **Commonly Confused With:** AWS Lambda - EC2 provides persistent virtual machines suitable for sustained workloads; Lambda runs short-lived functions in response to events without requiring server management.

AWS Inferentia and AWS Trainium

Category: Compute (Supplemental) **Exam Priority:** Not Officially In-Scope **Domain(s):** Not officially mapped **What It Is (Plain English):** AWS Inferentia and AWS Trainium are custom AWS-designed computer chips built specifically to accelerate ML inference (Inferentia) and ML

training (Trainium) at lower cost than general-purpose hardware. **Everyday Analogy:** These chips are like specialized power tools built for one job only, versus a general-purpose toolkit; they're less flexible, but much more efficient at exactly what they're designed for. **Primary Use Case:** Organizations running large-scale model training or inference use Trainium or Inferentia-based EC2 instances to reduce compute costs. **Why the AI Practitioner Needs to Know It:** Not explicitly named on the official AIF-C01 in-scope services list. Included here only because they're commonly mentioned in AWS AI infrastructure discussions; do not expect detailed exam questions about these specific chips. **Business Scenario:** A company running large-scale model training wanted to reduce their compute costs without sacrificing performance. They used EC2 instances powered by AWS Trainium chips for training. The result was lower training costs compared to general-purpose GPU instances. (Illustrative example only.) **Exam Clue Phrases:** - Not applicable for AIF-C01. **Commonly Confused With:** Amazon EC2 (GPU instances) - Inferentia and Trainium are specific custom-chip-based EC2 instance families optimized for AI workloads, while general EC2 GPU instances use third-party hardware.

SECTION 8: ADDITIONAL OFFICIALLY IN-SCOPE SERVICES (Storage, Networking, Containers, and Cost Management)

Amazon S3

Category: Storage **Exam Priority:** High **Domain(s):** Domain 1, Domain 3, Domain 4, Domain 5 **What It Is (Plain English):** Amazon S3 is the foundational AWS object storage service used to store training datasets, model artifacts, logs, RAG knowledge base documents, and any other files needed throughout the AI/ML lifecycle. **Everyday Analogy:** S3 is like an enormous, infinitely expandable digital filing cabinet that can hold virtually any type of file, accessible from anywhere. **Primary Use Case:** Almost every AI/ML workflow starts and ends with S3: raw data, processed data, model outputs, and documents for RAG knowledge bases all typically live there. **Why the AI Practitioner Needs to Know It:** S3 is referenced across all five domains, most often as the place where source data for training or RAG knowledge bases is stored. **Business Scenario:** A company building a RAG-based chatbot needed a central place to store the thousands of internal documents the chatbot would draw from. They used Amazon S3 to store those source documents before they were indexed into a knowledge base. The result was a reliable, scalable storage foundation for the entire RAG pipeline. **Exam Clue Phrases:** - "Store training data or documents" - "Source for a knowledge base" - "Object storage for model artifacts" - "Where does the data live before processing?" **Commonly Confused With:** Amazon S3 Glacier - S3 Standard is for frequently accessed data needing immediate retrieval; S3 Glacier is for archival data that is rarely accessed and can tolerate retrieval delays.

Amazon S3 Glacier

Category: Storage **Exam Priority:** Low **Domain(s):** Domain 5 **What It Is (Plain English):** Amazon S3 Glacier is a low-cost archival storage tier within Amazon S3 designed for data that must be retained for compliance or long-term reference but is rarely accessed. **Everyday Analogy:** Glacier is like an off-site storage unit for old records you're required to keep but rarely need to open. **Primary Use Case:** Organizations use Glacier to retain old training data, logs, or compliance

records cost-effectively over long periods. **Why the AI Practitioner Needs to Know It:** Know Glacier as the answer whenever a scenario emphasizes long-term retention and low cost over fast access, often tied to compliance and data retention policy in Domain 5. **Business Scenario:** A company needed to retain years of AI model audit logs for regulatory compliance, even though those logs were almost never accessed. They used Amazon S3 Glacier to store that historical data cost-effectively. The result was significant storage cost savings compared to keeping everything in standard S3. **Exam Clue Phrases:** - “Long-term archival storage” - “Rarely accessed data retained for compliance” - “Low-cost storage tier” **Commonly Confused With:** Amazon S3 Standard - Glacier trades immediate availability for significantly lower storage cost; S3 Standard is priced for data that must be retrieved quickly and frequently.

Amazon CloudFront

Category: Networking and Content Delivery **Exam Priority:** Low **Domain(s):** Domain 5 **What It Is (Plain English):** Amazon CloudFront is a content delivery network (CDN) that distributes web content, API responses, and AI application outputs to users globally through edge locations for reduced latency. **Everyday Analogy:** CloudFront is like having local warehouses near every customer so deliveries arrive faster, instead of shipping everything from one central location every time. **Primary Use Case:** Organizations use CloudFront to deliver AI application content quickly to users around the world. **Why the AI Practitioner Needs to Know It:** Know CloudFront as the network-edge delivery layer for end users, distinct from VPC, which is the private backend network. **Business Scenario:** A global company wanted users in different continents to experience low-latency responses from their AI-powered web application. They used Amazon CloudFront to deliver content from edge locations close to each user. The result was a faster, more consistent experience worldwide. **Exam Clue Phrases:** - “Content delivery network” - “Reduce latency for global users” - “Edge locations” **Commonly Confused With:** Amazon VPC - CloudFront delivers content to end users at the network edge; VPC is the private internal network where backend AWS resources like databases and servers reside.

Amazon VPC

Category: Networking and Content Delivery **Exam Priority:** Low **Domain(s):** Domain 5 **What It Is (Plain English):** Amazon VPC is a logically isolated private virtual network within AWS where customers deploy resources with full control over IP addressing, subnets, routing, and security, keeping AI workloads off the public internet. **Everyday Analogy:** VPC is like a private, gated neighborhood for your AWS resources, where you control exactly who can come and go. **Primary Use Case:** Organizations use VPC to isolate sensitive AI workloads and data from public internet exposure. **Why the AI Practitioner Needs to Know It:** Know VPC as the private network foundation that AWS PrivateLink builds on for secure connectivity, both tested in Domain 5. **Business Scenario:** A financial services company wanted to ensure their AI model training infrastructure was never exposed to the public internet. They deployed all of that infrastructure within an Amazon VPC. The result was a strong network isolation boundary protecting sensitive training data and models. **Exam Clue Phrases:** - “Private isolated network” - “Keep workloads off the public internet” - “Full control over networking and security” **Commonly Confused With:** AWS PrivateLink - VPC is the overall private network environment; PrivateLink

is a specific feature enabling private connectivity between VPCs and AWS services without traffic ever crossing the public internet.

Amazon ECS

Category: Containers **Exam Priority:** Low **Domain(s):** Domain 3 **What It Is (Plain English):** Amazon ECS is a fully managed container orchestration service that runs and scales Docker containers in AWS without requiring customers to manage the underlying cluster infrastructure. **Everyday Analogy:** ECS is like a moving company that not only transports your boxes (containers) but also manages the warehouse they're stored in, with AWS's own proprietary system. **Primary Use Case:** Organizations running containerized AI applications use ECS as a simpler, AWS-native alternative to Kubernetes. **Why the AI Practitioner Needs to Know It:** Know ECS as the simpler AWS-native container option, contrasted with EKS for teams that specifically need Kubernetes. **Business Scenario:** A team deploying a containerized custom inference application wanted a straightforward AWS-native solution without learning Kubernetes. They used Amazon ECS to run and scale their containers. The result was a simpler operational model with less infrastructure complexity to manage. **Exam Clue Phrases:** - "AWS-native container orchestration" - "Simpler container management, no Kubernetes required" **Commonly Confused With:** Amazon EKS - ECS is AWS's own native container management service with a simpler learning curve; EKS is the managed Kubernetes service for organizations that require Kubernetes-compatible tooling.

Amazon EKS

Category: Containers **Exam Priority:** Low **Domain(s):** Domain 3 **What It Is (Plain English):** Amazon EKS is a managed Kubernetes service that runs Kubernetes container workloads in AWS without requiring installation or management of the Kubernetes control plane. **Everyday Analogy:** EKS is like getting a fully maintained version of a specific, industry-standard moving system (Kubernetes) rather than AWS's own proprietary one. **Primary Use Case:** Organizations that need Kubernetes specifically, often for portability across cloud providers, use EKS for their containerized AI applications. **Why the AI Practitioner Needs to Know It:** Know EKS as the Kubernetes-specific option, used when an organization needs Kubernetes-native tooling or portability rather than AWS's own ECS system. **Business Scenario:** A company running AI workloads across multiple cloud providers wanted to use the same container orchestration tooling everywhere. They used Amazon EKS to run their Kubernetes-based deployments on AWS. The result was consistent tooling and portability across their multi-cloud environment. **Exam Clue Phrases:** - "Managed Kubernetes service" - "Kubernetes-native tooling or portability requirements" **Commonly Confused With:** Amazon ECS - EKS is for Kubernetes workloads requiring portability or Kubernetes-native tooling; ECS is AWS's proprietary service that is simpler to configure for AWS-first environments.

AWS Budgets

Category: Cloud Financial Management **Exam Priority:** Low **Domain(s):** Domain 2, Domain 5 **What It Is (Plain English):** AWS Budgets lets you set custom cost and usage thresholds for AWS spending, with automated alerts when actual or forecasted costs exceed defined budgets. **Everyday Analogy:** Budgets is like setting a spending limit alert on a credit card, so you're notified before you go over, rather than finding out at the end of the month. **Primary Use Case:** Organizations use Budgets to control unpredictable GenAI token-based costs by setting spending thresholds and getting early warnings. **Why the AI Practitioner Needs to Know It:** Know Budgets as the forward-looking, threshold-and-alert tool, contrasted with Cost Explorer's backward-looking analysis. **Business Scenario:** A startup using a foundation model through Bedrock was worried about unpredictable token-based costs as usage scaled. They used AWS Budgets to set spending thresholds with automated alerts. The result was early warning before costs grew out of control. **Exam Clue Phrases:** - "Set spending thresholds" - "Alerts when costs exceed a budget" - "Forward-looking cost control" **Commonly Confused With:** AWS Cost Explorer - Budgets sets spending thresholds and sends alerts when they are approached or exceeded; Cost Explorer provides historical analysis and visualizations of past spending.

AWS Cost Explorer

Category: Cloud Financial Management **Exam Priority:** Low **Domain(s):** Domain 2, Domain 5 **What It Is (Plain English):** AWS Cost Explorer is a tool for visualizing, analyzing, and understanding AWS costs and usage over time, helping organizations identify patterns, anomalies, and savings opportunities. **Everyday Analogy:** Cost Explorer is like reviewing your past bank statements to understand spending patterns, rather than setting a future spending alert. **Primary Use Case:** Organizations use Cost Explorer to analyze historical AI/ML and GenAI spending and identify where costs can be optimized. **Why the AI Practitioner Needs to Know It:** Know Cost Explorer as the backward-looking analysis tool, contrasted with Budgets' forward-looking alerts. **Business Scenario:** A company wanted to understand why their token-based GenAI costs had grown significantly over the past quarter. They used AWS Cost Explorer to analyze historical usage patterns and pinpoint the cause. The result was identification of a specific application driving most of the cost increase, allowing them to optimize it. **Exam Clue Phrases:** - "Analyze historical costs and usage" - "Identify spending patterns or anomalies" - "Understand past AWS spending" **Commonly Confused With:** AWS Budgets - Cost Explorer is used to analyze and understand historical costs; Budgets is used to set forward-looking spending limits and get notified when they are at risk of being exceeded.

SUMMARY TABLE 1: Services by Domain

(Officially in-scope services only)

Service	Domain 1	Domain 2	Domain 3	Domain 4	Domain 5
Amazon SageMaker AI					
Amazon SageMaker JumpStart					

Service	Domain 1	Domain 2	Domain 3	Domain 4	Domain 5
Amazon Bedrock					
Amazon Bedrock AgentCore					
Amazon Nova					
Amazon A2I					
Amazon Comprehend					
Amazon Kendra					
Amazon Lex					
Amazon Personalize					
Amazon Polly					
Amazon Rekognition					
Amazon Textract					
Amazon Transcribe					
Amazon Translate					
AWS Transform					
Amazon Q					
Kiro					
Strands Agents					
AWS Data Exchange					
Amazon EMR					
AWS Glue					
AWS Glue DataBrew					
AWS Lake Formation					
Amazon OpenSearch Service					
Amazon Quick					
Amazon Redshift					
AWS IAM					
Amazon Macie					
AWS KMS					
AWS Secrets Manager					
Amazon Inspector					
AWS Artifact					
AWS Audit Manager					
AWS CloudTrail					
Amazon CloudWatch					
AWS Config					
AWS Trusted Advisor					
AWS Well-Architected Tool					
AWS Lambda					
Amazon EC2					
Amazon Aurora					
Amazon RDS					
Amazon DynamoDB					
Amazon DocumentDB					
Amazon ElastiCache					
Amazon Neptune					
Amazon S3					

Service	Domain 1	Domain 2	Domain 3	Domain 4	Domain 5
Amazon S3 Glacier					
Amazon CloudFront					
Amazon VPC					
Amazon ECS					
Amazon EKS					
AWS Budgets					
AWS Cost Explorer					

SUMMARY TABLE 2: High-Priority Services Quick Reference
(High-priority, officially in-scope services only - your top study focus)

Service	One-Line Purpose	Top Exam Clue Phrase	Confused With
Amazon Bedrock	Access and customize foundation models from multiple providers via one API	“No infrastructure to manage” for GenAI	Amazon SageMaker AI
Amazon Bedrock AgentCore	Managed runtime to deploy, secure, and scale AI agents in production	“Securely deploy agents at scale”	Strands Agents
Amazon SageMaker AI	Build, train, and deploy custom ML models end to end	“Build a custom model from scratch”	Amazon Bedrock
Amazon SageMaker JumpStart	Pre-built model templates accessed through SageMaker	“Pre-built templates within SageMaker”	Amazon Bedrock
Amazon Nova	Amazon’s own family of foundation models within Bedrock	“Amazon’s own foundation models”	Amazon Bedrock (platform)
Amazon A2I	Routes low-confidence ML predictions to human reviewers	“Human-in-the-loop workflow”	SageMaker Model Monitor
Amazon Comprehend	Analyzes text for sentiment, entities, and key phrases	“Detect sentiment in text”	Amazon Textract
Amazon Lex	Builds custom conversational chatbots and voice bots	“Build a custom chatbot”	Amazon Polly
Amazon Polly	Converts written text into spoken audio	“Text-to-speech”	Amazon Transcribe

Service	One-Line Purpose	Top Exam Clue Phrase	Confused With
Amazon Rekognition	Analyzes images and video for objects, faces, and scenes	“Detect objects or faces in images”	Amazon Textract
Amazon Transcribe	Converts spoken audio into written text	“Speech-to-text”	Amazon Polly
Amazon Translate	Translates text between languages	“Translate content into another language”	Amazon Comprehend
Amazon Q	AI assistant for developers, IT pros, and business users	“AI assistant for AWS questions”	Amazon Lex
Amazon OpenSearch Service	Stores and searches vector embeddings for RAG	“Vector database for RAG”	Amazon Kendra
AWS IAM	Controls who can access which AWS resources	“Least privilege access”	AWS Secrets Manager
Amazon Macie	Discovers and classifies sensitive data in S3	“Discover sensitive data in S3”	Amazon Inspector
AWS Artifact	Self-service portal for downloading AWS compliance reports	“Download AWS compliance reports”	AWS Audit Manager
AWS Audit Manager	Continuously collects evidence of your own compliance	“Continuously collect compliance evidence”	AWS Artifact
AWS CloudTrail	Logs all API activity in an AWS account for auditing	“Who made this API call?”	Amazon CloudWatch
AWS Config	Tracks resource configuration state and compliance over time	“Detect configuration drift”	AWS CloudTrail
Amazon S3	Foundational object storage for data, models, and documents	“Store training data or documents”	Amazon S3 Glacier
Amazon Aurora	High-performance relational database with pgvector for RAG	“pgvector support for RAG”	Amazon RDS

End of Task 7 - AWS Service Dictionary Reference this document throughout your study program

and again in the final days before your exam. Next: Task 9 - Master Glossary (building on the service names introduced here)